Module 2: Fundamental data analysis

Physics B gives learners many opportunities to analyse data collected in practical sessions or provided for them. Learners should be exposed and trained in the techniques of analysis and the handling of experimental uncertainties throughout the course. In particular, the early parts of the course give learners the chance to practise the handling and analysing data. The Physics B approach to data analysis builds on the practical skills in module **1.1** as detailed below.

	Learning outcomes	Additional guidance
(a)	*Describe and explain:*	
	(i) factors affecting accuracy and uncertainty of measurements	
	(ii) the importance of recognising the largest source of uncertainty in a measurement.	
(b)	*Make appropriate use of:*	
	(i) SI units and their prefixes, standard form, angles in degrees and radians	Learners may be required to convert degrees to radians and *vice versa M0.1, M0.6, M4.7*
	(ii) the terms: accuracy, precision, resolution, sensitivity, response time, uncertainty, systematic error including zero error	as discussed in *The Language of Measurement* (ASE 2010)
	by sketching and interpreting:	
	(iii) simple plots of the distribution of measured values to estimate the mean (or median) value and the spread and to identifying potential outlying values and to suggest reasons to account for them	use of 'dot-plots' to estimate uncertainty
	(iv) a variety of different kinds of graphical plot and use of uncertainty bars to help establish the validity of measured data.	Including scatter graphs, pie-charts, log graphs
(c)	*Make calculations and estimates involving:*	
	(i) uncertainty of experimental data, the mean of results, range, spread and percentage uncertainties. Estimate of best fit gradients and intercepts with uncertainty	*M0.3, M1.2*
	(ii) estimating uncertainties when data are combined by addition, subtraction, multiplication, division and raising to powers	As set out in the ASE publication *Signs, Symbols and Systematics* (*The ASE Companion to 16–19 Science*, 2000). A rigorous statistical treatment is not required; learners will be expected to re-calculate values of the required quantity using extreme values of the variable(s) with the greatest uncertainty *M1.5*
	(iii) estimated magnitudes of everyday quantities.	